

TOPICAL REVIEW

Do multiple ionic interactions contribute to skeletal muscle fatigue?

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During intense exercise or electrical stimulation of skeletal muscle the concentrations of several ions change simultaneously in interstitial, transverse tubular and intracellular compartments. Consequently the functional effects of multiple ionic changes need to be considered together. A diminished transsarcolemmal K^+ gradient *per se* can reduce maximal force in non-fatigued muscle suggesting that K^+ causes fatigue. However, this effect requires extremely large, although physiological, K^+ shifts. In contrast, moderate elevations of extracellular $[K^+]_o$ ($[K^+]_o$) potentiate submaximal contractions, enhance local blood flow and influence afferent feedback to assist exercise performance. Changed transsarcolemmal Na^+ , Ca^{2+} , Cl^- and H^+ gradients are insufficient by themselves to cause much fatigue but each ion can interact with K^+ effects. Lowered Na^+ , Ca^{2+} and Cl^- gradients further impair force by modulating the peak tetanic force– $[K^+]_o$ and peak tetanic force–resting membrane potential relationships. In contrast, raised $[Ca^{2+}]_o$, acidosis and reduced Cl^- conductance during late fatigue provide resistance against K^+ -induced force depression. The detrimental effects of K^+ are exacerbated by metabolic changes such as lowered $[ATP]_i$, depleted carbohydrate, and possibly reactive oxygen species. We hypothesize that during high-intensity exercise a rundown of the transsarcolemmal K^+ gradient is the dominant cellular process around which interactions with other ions and metabolites occur, thereby contributing to fatigue.

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Repeated excitation of muscle leads to a reduced force production commonly known as fatigue. Several major mechanisms of fatigue are postulated, with contributions from altered motor drive, metabolites, reactive oxygen species and ions (for reviews see Enoka & Stuart, 1992; Fitts, 1994; Clausen, 2003; Lindinger, 2005; Allen *et al.* 2008). Many reviews that have considered the role of ions have focused on a single entity such as K^+ (Sejersted & Sjøgaard, 2000) or H^+ (Cairns, 2006). However, these reviews also indicate that during exercise multiple ionic changes occur across the sarcolemma, i.e. surface and transverse (t-) tubular membranes. The ionic change attracting most interest involves raised extracellular $[K^+]_o$ ($[K^+]_o$), and this has culminated in the K^+ hypothesis of fatigue: K^+ efflux from working muscle causes a rundown of the transsarcolemmal K^+ gradient sufficient to diminish force production. The $[K^+]_o$ is traditionally measured in venous plasma during exercise where values around 8 mM are regarded as extreme (Sejersted & Sjøgaard,

2000). However, exposure to these $[K^+]_o$ only moderately reduces maximum force in isolated non-fatigued muscle, and hence may not cause much fatigue. To counter this argument it is frequently postulated that the $[K^+]_o$ in the interstitial or t-tubular compartments of working muscle exceeds that in plasma (Jones & Bigland-Ritchie, 1986; Fitts, 1994; Sejersted & Sjøgaard, 2000). Furthermore, lowered extracellular $[Na^+]_o$ ($[Na^+]_o$) acts synergistically with raised $[K^+]_o$ to depress muscle force (Bouclin *et al.* 1995; Overgaard *et al.* 1997), and increased intra- and extracellular $[H^+]$ restores force at raised $[K^+]_o$ (Nielsen *et al.* 2001; Pedersen *et al.* 2004; Kristensen *et al.* 2005). These interactions make it unlikely that K^+ acts in isolation during fatigue. Therefore, the purpose of this review is to highlight how the multiple ionic changes that occur during exercise interact in a manner that contributes to fatigue of mammalian/human muscles under physiological conditions.

Table 1. Representative ionic changes in various muscle compartments during high-intensity exercise in humans

Compartment	[K ⁺] (mM)	[Na ⁺] (mM)	[Ca ²⁺] (mM)	[Cl ⁻] (mM)	pH (pH units)
Venous plasma					
Rest	4.1 (3.8–4.4)	140 (139–144)	1.22	104.5 (104, 105)	7.41 (7.38–7.44)
Exercise	5.8 (4.7–6.2) ref. 1,3,4,7,8	149 (145–156) ref. 1,7,9	1.28 ref. 7	105 (105, 105) ref. 1,2	7.15 (6.96–7.33) ref. 2,3,7,9
Interstitial					
Rest	4.5 (4.3–5.0)	142	—	95*	7.36 (7.34,7.38)
Exercise	10.5 (9.0–13.7) ref. 4,5,8,9	128 ref. 9	—	—	7.01 (6.98,7.04) ref. 6,9
Intracellular					
Rest	159 (150–168)	9 (6–13)	—	19 (15,22)	7.14 (7.08–7.17)
Exercise	131 (129–134) ref. 1,2,3	20 (16–24) ref. 1,2,3	—	27 (26,28) ref. 1,2	6.71 (6.64–6.80) ref. 1,2,3

Values are the average of mean data from the studies cited (range of mean data is in parentheses). Data were obtained from various leg muscles of male subjects before and immediately after intense cycling, rowing, incremental one-legged knee extensions or plantar flexions of 5–20 min duration. Interstitial data were obtained using microdialysis and intracellular data from muscle biopsies. References: 1, Bergström *et al.* (1971); 2, Sahlin *et al.* (1978); 3, Sjøgaard *et al.* (1985); 4, Green *et al.* (2000); 5, Juel *et al.* (2000); 6, Street *et al.* (2001); 7, Nielsen *et al.* (2002); 8, Nielsen *et al.* (2003); 9, Street *et al.* (2005). *Predicted from a Donnan equilibrium with ions in plasma. — indicates values yet to be determined.

Ion shifts in working muscle

Exercise-induced ion shifts (fluxes which alter ion contents), water movements, physicochemical reactions,

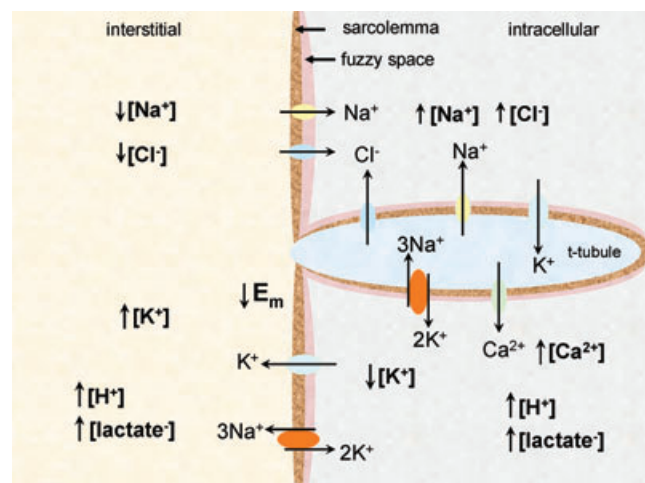


Figure 1. Schematic representation of transsarcolemmal ion fluxes and ion concentration changes in various muscle compartments (interstitial, t-tubular, intracellular) during electrical stimulation or exercise

Sarcolemma includes both surface and t-tubular membranes. Fuzzy space is a microdomain in the subsarcolemmal region. Pathways for ion fluxes include: K⁺ efflux via delayed rectifier K⁺, K_{ATP} and K_{Ca} channels; Na⁺ influx via voltage-dependent Na⁺ and stretch-activated channels; Ca²⁺ influx via L-type Ca²⁺, stretch-activated and store-operated channels; and Cl⁻ influx via CIC-1 channels. The Na⁺-K⁺-pump (depicted) acts to maintain Na⁺ and K⁺ gradients. Ionic changes are also predicted for the t-tubular lumen (see text).

and metabolic processes lead to ion concentration changes in compartments proximal to the sarcolemma (Fig. 1). It is consistently observed that multiple ionic changes (K⁺, Na⁺, Ca²⁺, Cl⁻, H⁺, HCO₃⁻, Mg²⁺, H₂PO₄⁻, PCr²⁻, lactate⁻) occur in the various compartments with intense exercise or electrical stimulation (Fig. 1; Table 1; Sembrowich *et al.* 1983; Lindinger & Heigenhauser, 1988, 1991). We focus on the first five inorganic ion species where the tendency is for a loss of muscle K⁺, and gain of muscle Na⁺, Cl⁻, Ca²⁺ and water, whilst H⁺ is elevated both in muscle and interstitial fluid/plasma. The magnitude of these ion shifts inevitably depends on the fatigue model employed, i.e. the exercise or stimulation regime, preparation type, and muscle environmental conditions (Cairns *et al.* 2005). We now describe the ion concentrations measured in interstitial and intracellular compartments, and estimated changes for the t-tubules since together they determine directly the resting membrane potential (*E_m*), ionic currents and sarcolemmal excitability.

Interstitial compartment. All ion concentrations in the interstitium are likely to be spatially graded, from capillary to sarcolemma, with the magnitude and direction of the gradients dependent on muscle contractile activity, arterial and venous capillary ion concentrations (diffusion limitation) and flow rate through the capillary (perfusion limitation). Table 1 shows that interstitial ion concentrations are different from those in venous plasma both at rest and during exercise, confirming that plasma

does not represent the sarcolemmal milieu. Thirty years ago it was shown, with ion-sensitive microelectrodes, that interstitial $[K^+]_i$ ($[K^+]_i$) increased to 8–15 mM during prolonged tetani in animal muscle (Hník *et al.* 1976) or sustained voluntary contractions in human muscle (Vyskočil *et al.* 1983). However, these $[K^+]_i$ values have only recently gained widespread acceptance following confirmation with the microdialysis technique (Table 1). Microdialysis studies (Green *et al.* 2000; Juel *et al.* 2000; Nielsen *et al.* 2003; Street *et al.* 2005) revealed that $[K^+]_i$ increases linearly with workload up to 9–14 mM. Maximal $[K^+]_i$ can differ by up to 6 mM when measured using several probes inserted within a muscle during the same exercise bout. This may arise because probes are located adjacent to different fibre types, since there is greater K^+ efflux per action potential in fast- than slow-twitch fibres (Sejersted & Sjøgaard, 2000; Clausen *et al.* 2004). Consistent with these findings, the $[K^+]_i$ in the interfibre space of isolated whole mouse muscle increases from 5 to 10 mM during severe fatigue with repeated brief tetani (Juel, 1986) – a stimulation regime used to represent locomotion (Cairns *et al.* 2005). Surprisingly, such raised $[K^+]_i$ occur despite net water flux from plasma to interstitium (Sjøgaard *et al.* 1985; Juel, 1986; Lindinger *et al.* 1987). Table 1 also illustrates that during intense exercise $[Na^+]_i$ and pH_i decline. The estimated resting $[Cl^-]_i$ is marginally lower than plasma $[Cl^-]$ due to a Donnan equilibrium. Moreover, both $[Cl^-]_i$ and $[Ca^{2+}]_i$ should fall during exercise, by ~10% with the greater interstitial fluid volume, and to a greater extent with Cl^- or Ca^{2+} shifts into muscle. Clearly, multiple and large changes of interstitial ions occur during heavy exercise.

t-Tubular compartment. A popular hypothesis is that ionic changes are even greater in the t-tubules than interstitium due to diffusion limitations (Jones & Bigland-Ritchie, 1986; Fitts, 1994; Sejersted & Sjøgaard, 2000). The basis for this hypothesis is that the t-tubular membranes constitute a major portion of the sarcolemmal area, engulf a tiny lumen, follow a branched tortuous pathway, and contain numerous ion channels and transporters which permit ion fluxes (Fig. 1; Jurkat-Rott *et al.* 2006; Stephenson, 2006). Convincing evidence now supports restricted diffusion of K^+ (Swift *et al.* 2006; Shorten & Soboleva, 2007), Na^+ (Fujishiro & Kawata, 1992) and Ca^{2+} (Friedrich *et al.* 2001) in the t-tubular lumen. Ion concentrations have not been measured in the t-tubular fluid – using microelectrodes or microdialysis probes – although mathematical modelling has been used to calculate values. First, the estimated t-tubular $[K^+]_i$ ($[K^+]_{t-sys}$) for a single tetanus (40 Hz) increases from 4–6 mM at rest to 9–14 mM at 1 s stimulation, although it depends on the fibre diameter (Wallinga *et al.* 1999; Shorten & Soboleva, 2007). Also the modelled $[K^+]_{t-sys}$

gradients across a 35 μ m diameter fibre indicate a sharp elevation to 10 mM within 5 μ m of the surface, then a gradual rise towards the centre (Shorten & Soboleva, 2007). However, these analyses probably underestimate $[K^+]_{t-sys}$ during repeated contractions. Second, the predicted $[Na^+]_{t-sys}$ falls by a greater extent with increasing stimulation frequency in amphibian muscle: from 120 to 65 mM with 60 Hz stimulation for 2 s (Bezanilla *et al.* 1972). Also reducing $[Na^+]_o$ by one-half impairs action potential propagation in t-tubular membranes during a 2 s tetanus (Bezanilla *et al.* 1972; Duty & Allen, 1994). Third, t-tubular membranes have several pathways for Ca^{2+} influx (Jurkat-Rott *et al.* 2006; Launikonis & Rios, 2007). Modelling or measurement using fluorescence indicators within sealed t-tubules indicate that $[Ca^{2+}]_{t-sys}$ falls considerably during action potentials (Launikonis *et al.* 2007) or with sustained depolarization (Friedrich *et al.* 2001; Launikonis & Rios, 2007). In contrast, radioisotope studies imply $[Ca^{2+}]_{t-sys}$ may increase with repetitive stimulation (Bianchi & Narayan, 1982) but direct measurements have yet to be made during fatigue. Fourth, t-tubular membranes also possess a considerable Cl^- conductance (g_{Cl}) (Dulhunty, 1979; Dutka *et al.* 2008), which allows Cl^- influx during action potentials or depolarization (McCaig & Leader, 1984; Heiny *et al.* 1990), making it plausible that $[Cl^-]_{t-sys}$ declines during fatigue. Taken together, and based on the calculated ion concentration changes in the t-tubules exceeding those in the interstitium, interactive effects amongst ions in the t-tubules are likely to contribute to fatigue. However, direct measures of such t-tubular changes during fatigue are needed to further our understanding.

Intracellular compartment. Table 1 shows the changes of intracellular ion concentrations after 5–20 min of heavy exercise, which include a lowered intracellular $[K^+]_i$ ($[K^+]_i$), raised $[Na^+]_i$, which may double, raised $[Cl^-]_i$, lowered pH_i , and also increased $[lactate^-]_i$ and lowered $[HCO_3^-]_i$, with diminished glycogen content (Sahlin *et al.* 1978; Sjøgaard *et al.* 1985). The ionic changes are smaller with brief bouts of intense exercise, when the Na^+ and Cl^- contents increase but concentrations may remain unchanged due to increased intracellular fluid volume (Kowalchuk *et al.* 1988; Lindinger *et al.* 1995). Ionic changes likewise occur during prolonged submaximal exercise, but again are smaller than during intense exercise, with pH_i largely unaffected and glycogen depletion being greater (Ahlborg *et al.* 1967; Sjøgaard, 1983, 1986).

Repeated stimulation of animal muscle permits detailed analysis of intracellular ionic changes (Fig. 1). During severe fatigue (peak force < 50% initial) there is lowered $[K^+]_i$ and pH_i , and raised $[Na^+]_i$, $[Cl^-]_i$, resting $[Ca^{2+}]_i$ and water content (Sembrowich *et al.* 1983; Juel, 1986, 1988a; Sjøgaard, 1986; Lindinger & Heigenhauser, 1988,

1991; Clausen *et al.* 2004). Furthermore, these studies reveal greater ionic changes in fast- than slow-twitch fibres. Juel (1986) found that when fast-twitch extensor digitorum longus (EDL) and slow-twitch soleus muscles of mice were fatigued with repeated tetani, the $[K^+]_i$ fell from 182 mM in EDL or 168 mM in soleus to 130–135 mM. The final $[K^+]_i$ values were similar in both muscle-types and compatible with values in fatigued human muscle (Table 1). Moreover, with repeated isometric tetani $[Na^+]_i$ increases from ~13 to 21–23 mM (Fong *et al.* 1986; Juel, 1986) but can reach 45 mM (Lindinger *et al.* 2005), and such changes are exacerbated with eccentric contractions (McBride *et al.* 2000; Yeung *et al.* 2003). An elevated Cl^- content occurs in muscles stimulated for several minutes which tend to convert into raised $[Cl^-]_i$ only in fast-twitch fibres (Krnjević & Miledi, 1958; Sembrowich *et al.* 1983; Lindinger *et al.* 1987; Lindinger & Heigenhauser, 1988). A large intracellular acidosis also features in late fatigue with several different stimulation regimes (Sembrowich *et al.* 1983; Lindinger & Heigenhauser, 1988, 1991; Cairns, 2006). An increased resting $[Ca^{2+}]_i$ (Sembrowich *et al.* 1983; Lindinger & Heigenhauser, 1988; Allen *et al.* 2008) may result from Ca^{2+} influx in both fast- and slow-twitch muscles (Gissel & Clausen, 2000), and also altered Ca^{2+} handling by the sarcoplasmic reticulum (SR) (Allen *et al.* 2008). The intracellular ion concentrations most pertinent for excitability are presumably localized in the subsarcolemmal region where concentrations may differ to the spatially average intracellular concentrations. Indeed, there is mounting evidence of microdomains for $[Na^+]_i$, with the so-called fuzzy space (Fig. 1, Semb & Sejersted, 1996), as well as for both $[Ca^{2+}]_i$ (Stroffekova & Heiny, 1997) and $[ATP]_i$ (Dutka & Lamb, 2007b) in the triadic region. Although large changes of spatially averaged intracellular ion concentrations have been established (Table 1), the quantification of multiple ion concentrations in the subsarcolemmal region during severe fatigue is now required.

Resting membrane potential. The detrimental effects of K^+ during fatigue are thought to be mediated via sarcolemmal resting membrane potential (E_m) (Sejersted & Sjøgaard, 2000). Direct measurements of E_m require the use of microelectrodes, which limits these studies to fatiguing stimulation of animal muscles. The resting E_m (–90 to –70 mV) of mammalian muscle usually depolarizes by 7–25 mV during severe fatigue with repeated tetani (Juel, 1986, 1988a,b; Atrakchi *et al.* 1994; McBride *et al.* 2000; Karelis *et al.* 2005; Lindinger *et al.* 2005). However, the end-point E_m value varies with the starting E_m value, extracellular solution composition, stimulation regime and muscle fibre type. Importantly, it critically depends on how soon the electrode is inserted after stimulation ceases, since E_m recovers with a time

constant of ~1 min at 37°C (Juel, 1986). Indeed, when extrapolated back to the end of stimulation the E_m depolarizes to less than –60 mV in isolated mouse soleus and EDL muscles (Juel, 1986, 1988a) and –65 mV in rat muscle *in situ* (Lindinger *et al.* 2005). Notably, the E_m during a brief non-fatigued tetanus can depolarize to between –66 and –50 mV in the surface (Cairns *et al.* 2003; Gong *et al.* 2003) and t-tubular membranes (DiFranco *et al.* 2005). Presumably this depolarization is additive to the resting E_m measured after fatigue. Moreover, reports of hyperpolarization with repeated tetani (Fong *et al.* 1986; Hicks & McComas, 1989; Atrakchi *et al.* 1994) are not a conundrum since an early hyperpolarization (15 s) reverses into depolarization with more prolonged stimulation (300 s) (Lindinger *et al.* 2005). Hence the occurrence of depolarization may be linked to the severity of fatigue.

The E_m can be calculated for muscle fibres in exercising humans using the Goldman–Hodgkin–Katz (GHK) equation (described for 37°C):

$$E_m = 61.5 \log_{10} \left(\frac{[K^+]_i + \alpha[Na^+]_i + \gamma[Cl^-]_i}{[K^+]_o + \alpha[Na^+]_o + \gamma[Cl^-]_o} \right)$$

In human muscle α , the Na^+/K^+ permeability ratio, is 0.01 (Cunningham *et al.* 1971), and γ , the Cl^-/K^+ permeability ratio, is 3 (Kwieciński *et al.* 1984). The GHK equation shows that E_m is influenced by the $[K^+]_i/[K^+]_o$ ratio and hence K^+ equilibrium potential (E_K) (Sejersted & Sjøgaard, 2000). Notably, the E_K value obtained during fatiguing exercise in human muscle (calculated using data in Table 1) is similar to that in severe stimulation-induced fatigue in rodent muscles (Juel, 1986). Thus a large decline of K^+ gradient occurs with volitional contractions and does not require high-frequency stimulation. However, complexity arises in the calculation of E_m because it is influenced by other ion concentrations such as Na^+ and Cl^- , the sarcolemmal permeabilities (or conductances) for these ions may change during activity (Fink & Lüttgau, 1976; Fink *et al.* 1980), and a variable electrogenic Na^+-K^+ -pump current may occur (Clausen, 2003). Despite these cautions we calculated the E_m of fatigued human limb muscle using the measured ion concentrations from interstitial and intracellular spaces (Table 1). We acknowledge three assumptions: (i) a contribution from Cl^- to the E_m is likely to be small and can be ignored because of Cl^- influx over the 5–20 min exercise period (Bergström *et al.* 1971; Sahlin *et al.* 1978) which reduces the Cl^- equilibrium potential, and γ also falls since g_{Cl} declines with acidosis (Pedersen *et al.* 2005) and K^+ conductance increases (Fink & Lüttgau, 1976; Fink *et al.* 1980); (ii) α is regarded as being unchanged during fatigue as shown for metabolic exhaustion (Fink *et al.* 1980); and (iii) an electrogenic contribution of the Na^+-K^+ -pump during fatigue is not included because of

uncertainty about how the pump contribution changes (Clausen, 2003; Aughey *et al.* 2005; McKenna *et al.* 2006). The calculated E_m for resting muscle of -88 mV was similar to measured E_m values *in vivo* (Cunningham *et al.* 1971). During fatigue when $[K^+]_i$ is 9 mM the calculated E_m was -68 mV, and when $[K^+]_i$ is 14 mM the E_m was -57 mV. This E_m range for fatigued human muscle is in agreement with values measured during severe fatigue in rodent muscle (see above). Application of GHK analysis to ionic data from fatigued rodent muscle (Juel, 1986, 1988a; Sjøgaard, 1986; Lindinger & Heigenhauser, 1991) provides calculated E_m values of -70 to -55 mV. Importantly, this approach may underestimate the depolarization if larger ionic changes in the t-tubules influence the surface E_m or overestimate the depolarization if a greater electrogenic contribution from the Na^+-K^+ -pump occurs.

Summary. During muscle activity the simultaneous changes of intra- and extracellular $[K^+]$, $[Na^+]$ and $[Cl^-]$, ionic conductances (modified via $[Ca^{2+}]_i$ and $[H^+]_i$), and Na^+-K^+ -pump function, all influence the E_m (and time course of changes in E_m), excitability and ultimately contractile performance. These interactions are discussed in detail below.

Effects of potassium ions on muscle function

K^+ effects on contraction. The K^+ hypothesis of fatigue can be tested by exposing a non-fatigued muscle preparation to raised $[K^+]_o$ to determine whether contractility is impaired. The findings include no change, potentiation or depression of force depending on the concentration and duration of K^+ exposure, and whether submaximal/maximal contractions are tested (Fig. 2A). The steady-state peak tetanic force– $[K^+]_o$ relationship has been established for the entire physiological range of $[K^+]_o$ in rat soleus muscle (Cairns *et al.* 1995), and mouse fast- and slow-twitch muscle (Cairns *et al.* 1997). The relationship in fast-twitch muscle (Fig. 2A) shows that the tetanus was hardly affected when $[K^+]_o$ was increased from 4 to 7 mM, fell by $\sim 25\%$ at 10 mM, then declined abruptly to complete suppression at 12 mM. Surprisingly, slow-twitch muscle showed greater tetanus depression over this $[K^+]_o$ range (Clausen & Everts, 1991; Cairns *et al.* 1997; Hansen *et al.* 2005). The incomplete force suppression seen at 14–20 mM $[K^+]_o$ when using long stimulation pulses (Cairns *et al.* 1995, 1997) is an unphysiological response because these pulses trigger some Ca^{2+} release independently of action potentials (Cairns *et al.* 2007). Many studies now confirm that tetanus depression is a feature of 7–14 mM $[K^+]_o$ in both mammalian and amphibian muscles (Juel, 1988a; Renaud & Light, 1992; Overgaard *et al.* 1997). When using mechanically skinned mammalian fast-twitch fibres,

the $[K^+]_i$ can be lowered to reduce the K^+ gradient across t-tubular membranes and this manipulation also diminishes force (Ørtenblad & Stephenson, 2003; Pedersen *et al.* 2004; Dutka & Lamb, 2007a). Together

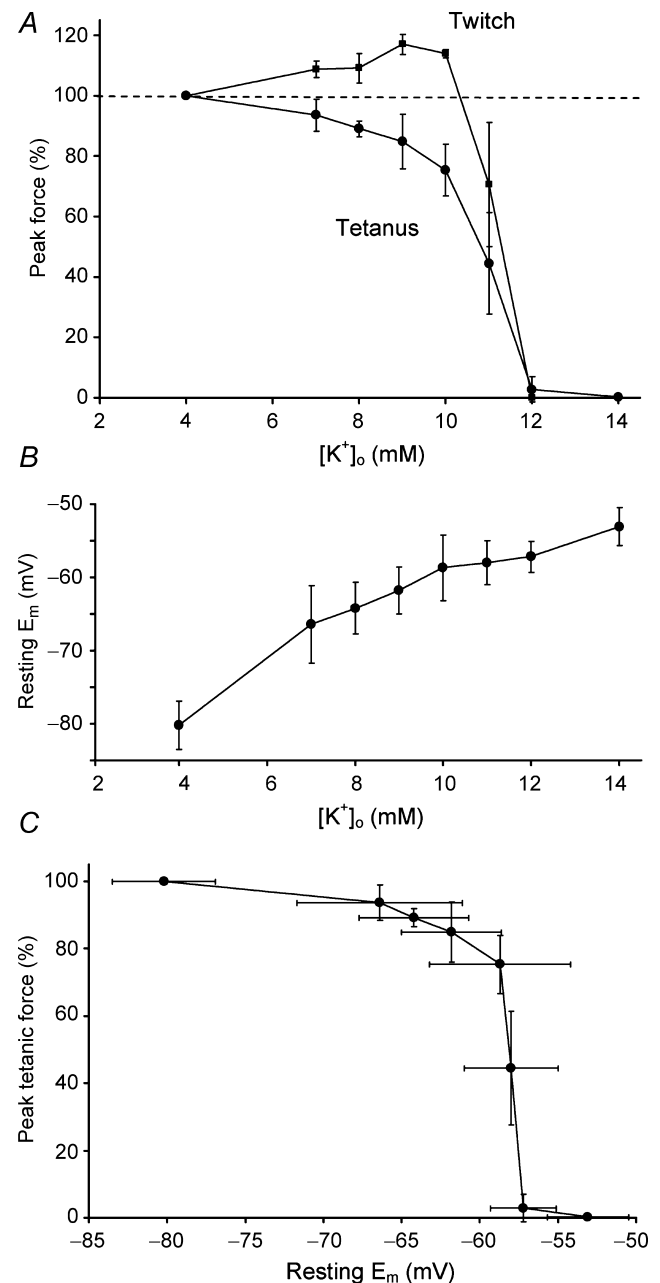


Figure 2. The influence of reduced transarcolemmal K^+ gradient on contractile force is mediated via depolarization of the resting E_m in isolated mouse extensor digitorum longus (EDL) muscle. **A**, the steady-state peak force– $[K^+]_o$ relationships for twitch (■) and tetanic (●) contractions. Tetani were evoked at 200 Hz for 1 s, with supramaximal (20 V, 0.1 ms) pulses delivered via parallel plate electrodes, 25°C. **B**, the resting E_m – $[K^+]_o$ relationship – determined using surface fibres bathed for 50–80 min at each $[K^+]_o$. **C**, the peak tetanic force–resting E_m relationship derived using the data in **A** and **B**. All data points are mean values $\pm$ s.d. From Cairns *et al.* (1997) (modified with permission) with inclusion of further experiments.

these studies indicate that a very large decline of the K^+ gradient has the potential to cause severe fatigue.

One observation which questions the K^+ hypothesis is the slow time course (i.e. > 60 min) from when $[K^+]_o$ is raised around whole muscles until a steady force decline is achieved (Clausen & Everts, 1991; Cairns *et al.* 1995, 1997, 2004). This can be explained by at least three processes: (i) diffusional delays occur in isolated, superfused whole muscles since the use of smaller fibre-bundles abbreviates the time for the full effects of K^+ (Cairns *et al.* 1995); (ii) changes of interfibre $[K^+]_o$ in the centre of muscle occur slowly because of regulation by Na^+-K^+ -pumps (Clausen & Everts, 1991); (iii) low $[Cl^-]_o$ accelerates the depressive effects of K^+ in whole muscles (Cairns *et al.* 2004), which implies that a normal $[Cl^-]_o$ gives protection. Importantly, the K^+ -induced force decline in fibre-bundles is complete in 5–20 min (Cairns *et al.* 1995), which is similar to the time until cessation of intense exercise (Table 1).

Another test of the K^+ hypothesis involves changing $[K^+]_o$ prior to fatiguing stimulation then investigating effects on fatigue kinetics: raised $[K^+]_o$ is predicted to accelerate fatigue by giving rundown of the K^+ gradient a head start. Indeed, incubation at 10 mM $[K^+]_o$ increases the rate of force loss during prolonged tetani (Cairns & Dulhunty, 1995; Clausen & Nielsen, 2007) and 7 mM $[K^+]_o$ moderately accelerates late fatigue during repeated tetani in mouse soleus (Cairns, 2005). Conversely, when $[K^+]_o$ was lowered to 2 mM, intermittent tetanic fatigue was slower (Cairns, 2005). These results support the K^+ hypothesis but could also be explained if K^+ interacted with other factors that change during fatigue.

K^+ effects on E_m . The detrimental effects of a reduced K^+ gradient are thought to be mediated entirely via resting E_m (Hodgkin & Horowitz, 1959; Cairns *et al.* 1997; Sejersted & Sjøgaard, 2000). The depolarization with raised $[K^+]_o$ is illustrated by the resting E_m – $[K^+]_o$ relationship (Fig. 2B) and was quantified by the GHK equation (Cairns *et al.* 1997). We then derived the peak tetanic force–resting E_m relationship (Fig. 2C) which we consider provides the best information to evaluate the contribution of K^+ to fatigue: it accounts for both raised $[K^+]_i$ and lowered $[K^+]_i$. This relationship shows a large safety margin with the E_m depolarizing to ~ -65 mV before there is any force loss. Further depolarization over -65 to -55 mV (the critical E_m range) leads to complete suppression of force. A $\sim 25\%$ force loss occurs between -65 and -60 mV, and a $\sim 75\%$ decrement between -60 and -55 mV. This relationship is similar in fast- and slow-twitch mouse muscles at 25°C (Cairns *et al.* 1997) and in rat soleus at 30°C (Cairns *et al.* 1995). When derived for mechanically skinned fibres using calculated E_m , the relationship has a similar form but is shifted leftwards by ~ 5 mV, so that at -65 mV the peak force is less than 50% of the initial force (Ørtenblad &

Stephenson, 2003; Pedersen *et al.* 2004); nevertheless a large variation occurs between individual fibres (Dutka & Lamb, 2007a). Moreover, species differences exist with force loss occurring over a 5–15 mV less depolarized E_m range in amphibian muscle (Renaud & Light, 1992). Clearly, it would be valuable to establish this relationship for human muscle. Since the E_m can be depolarized beyond -65 mV during severe fatigue in mammals and humans (see earlier section) this would be expected to cause extensive force loss (Fig. 2C).

Mechanism(s) for K^+ effects. Two main mechanisms account for the depolarization-induced impairment of contraction. The first mechanism involves effects on action potentials, which include a smaller amplitude, intermittent or complete failure during train stimulation, and complete unexcitability – as shown with either M-wave (compound muscle action potential) or intracellular recordings (Renaud & Light, 1992; Overgaard *et al.* 1999; Rich & Pinter, 2003; Pedersen *et al.* 2005). Such effects are likely to be mediated via a reduced sarcolemmal Na^+ conductance (g_{Na}) caused by lesser activation and/or greater slow- or fast-inactivation of Na^+ channels (Ruff, 1996; Rich & Pinter, 2003) and a reduced driving force for the transsarcolemmal Na^+ -current, i.e. the difference between the E_m and Na^+ equilibrium potential ($E_m - E_{Na}$). These changes contribute to a smaller Na^+ current (I_{Na}), since $I_{Na} = (E_m - E_{Na}) \times g_{Na}$, along with causing a prolonged refractory period (Dutka & Lamb, 2007a), increased threshold (Rich & Pinter, 2003), and slowed conduction velocity (Juel, 1988b) for action potentials. The second mechanism involves inactivation of the voltage sensors of excitation–contraction coupling, which is a feature of fast-twitch fibres (Chua & Dulhunty, 1988; Dutka & Lamb, 2007a). This process requires greater depolarization than for the impairment of excitability (Ørtenblad & Stephenson, 2003; Pedersen *et al.* 2004), so that voltage sensor inactivation is likely to be a minor contributor to fatigue. Importantly, impaired excitability is demonstrated as a fatigue mechanism with some stimulation regimes (Fitts, 1994; Cairns & Dulhunty, 1995; Gong *et al.* 2003; Clausen *et al.* 2004; Lindinger, 2005) thereby further supporting the K^+ hypothesis.

Summary. The depressive effect of physiological but large reductions of K^+ gradient on peak tetanic force in non-fatigued muscle preparations, the modulation of fatigue kinetics with small changes of $[K^+]_o$, the sarcolemmal depolarization induced with raised $[K^+]_o$, and associated impairment of excitability (as seen in some forms of fatigue), all support the hypothesis that large reductions of K^+ gradient contribute to fatigue.

Ion-ion interactions and muscle function

The observation that $[K^+]_i$ reaches 10–15 mM during sustained contractions without being accompanied by a large decline of force (Hník *et al.* 1976; Vyskočil *et al.* 1983) strongly challenges the K^+ hypothesis. However, in this situation the high $[K^+]_i$ may not have caused a sufficient depolarization due to protection by active Na^+ – K^+ -pumps (Clausen, 2003) or because other ions (e.g. Ca^{2+} , Cl^- , H^+) antagonized the K^+ effects (Cairns *et al.* 1998, 2004; Nielsen *et al.* 2001). In consequence, we now describe several ionic interactions and their potential mechanisms.

Sodium. The Na^+ gradient ($[Na^+]_i/[Na^+]_o$), and hence E_{Na} , often falls during intense exercise (Table 1) with its role in fatigue being tested by lowering $[Na^+]_o$ around non-fatigued muscle. The peak tetanic force– $[Na^+]_o$ relationship displays a large safety margin before force declines with lowered $[Na^+]_o$ in both fast- and slow-twitch mammalian (Jones & Bigland-Ritchie, 1986; Overgaard *et al.* 1997; Cairns *et al.* 2003) and amphibian muscles (Bezanilla *et al.* 1972; Bouclín *et al.* 1995). Thus, when $[Na^+]_o$ is reduced by one-half (equivalent to doubling $[Na^+]_i$ during exercise), the peak tetanic force falls by 10–15% (Jones & Bigland-Ritchie, 1986; Overgaard *et al.* 1997; Cairns *et al.* 2003). However, lowering $[Na^+]_o$ avoids the potential benefit of raised $[Na^+]_i$ to stimulate the Na^+ – K^+ -pump (Clausen, 2003; Lindinger *et al.* 2005). Hence, in most forms of exercise any decline of Na^+ gradient *per se* is unlikely to induce much fatigue. Lowered $[Na^+]_o$ has an impact on action potentials through a smaller amplitude, severe skipping or propagation failure during trains, and makes fibres completely unexcitable (Bezanilla *et al.* 1972; Duty & Allen, 1994; Cairns *et al.* 2003). Such effects do not involve altered E_m (Cairns *et al.* 2003) although extremely low $[Na^+]_o$ can cause depolarization through reduced Na^+ – K^+ -pump activity (Overgaard *et al.* 1997). Furthermore, pre-equilibration with reduced $[Na^+]_o$ exacerbates fatigue during either prolonged or repeated tetani (Jones & Bigland-Ritchie, 1986; Cairns & Dulhunty, 1995; Cairns *et al.* 2003), and these effects are more rapid and exceed those for K^+ effects on fatigue kinetics (Cairns & Dulhunty, 1995; Cairns, 2005). These findings support the proposal that a lowered Na^+ gradient may reduce force by interacting with other fatigue factors during stimulation.

Moderately raised $[K^+]_o$ (4 to 8 mM) and lowered $[Na^+]_o$ (147 to 100 mM) reduce peak tetanic force to 67% of control in mouse soleus muscle (Fig. 3). Individually these ionic changes exert minor effects, and the predicted additive effect (product of the individual effects) was to 85% control. Clearly, this Na^+ – K^+ interaction involves a synergist impairment of force. The lowered $[Na^+]_o$ shifts the peak tetanic force– $[K^+]_o$ relationship

leftwards to make the muscles more susceptible to raised $[K^+]_o$, as seen with other muscles (Bouclín *et al.* 1995; Overgaard *et al.* 1997, 1999). In skinned fibres, reduced $[K^+]_i$ with raised $[Na^+]_i$ led to greater force depression with doublet stimulation although the twitch benefited with enhanced t-tubular Na^+ – K^+ -pump activity mediated via raised $[Na^+]_i$ (Nielsen *et al.* 2004). A third test, with carbacholine, which opens acetylcholine receptor channels to increase $[Na^+]_i$ and lower $[K^+]_i$, also depressed peak tetanic force (Macdonald *et al.* 2005). The depressive interaction between lowered Na^+ and K^+ gradients involves reduced excitability (Overgaard *et al.* 1999) and can be explained by lowered $[Na^+]_o$ reducing the driving force (i.e. $E_{Na} - E_m$) for I_{Na} , which is a process also influenced by K^+ .

Calcium. The possibility that Ca^{2+} influx lowers $[Ca^{2+}]_{t-sys}$ to cause fatigue has been tested by exposing non-fatigued muscle to nominally Ca^{2+} -free solutions. This has little effect on force (Cairns *et al.* 1998; Zhao *et al.* 2005), which implies that severe Ca^{2+} depletion *per se* would not cause fatigue. However, lowered $[Ca^{2+}]_o$, or blockers of various Ca^{2+} entry pathways, exacerbates fatigue during repeated or continuous tetani (Williams & Ward, 1991; Cairns *et al.* 1998; Zhao *et al.* 2005; Ducret *et al.* 2006; Germinario *et al.* 2008) and raised $[Ca^{2+}]_o$ (5–10 mM), or availability of more Ca^{2+} entry

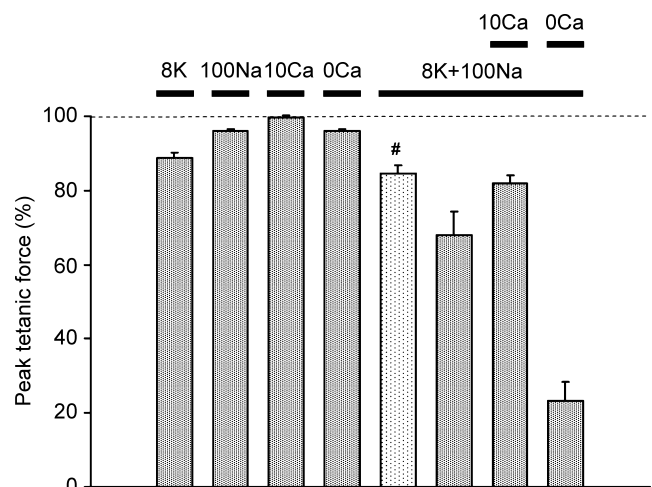


Figure 3. Interactive effects of raised $[K^+]_o$, lowered $[Na^+]_o$, and altered $[Ca^{2+}]_o$ on peak tetanic force in isolated mouse soleus muscle

Data are from Cairns *et al.* (1998). Each data point is the mean steady-state value ($\pm$ S.E.M.). Tetani were evoked at 125 Hz for 2 s, with supramaximal (20 V, 0.1 ms) pulses delivered via parallel plate electrodes, 25°C. The control Krebs solution included 4 mM K^+ , 147 mM Na^+ , and 1.3 mM Ca^{2+} . 8K – Krebs solution with 8 mM K^+ . 100Na – Krebs solution with 100 mM Na^+ . 10Ca – Krebs solution with 10 mM Ca^{2+} . 0Ca – Krebs solution that is nominally Ca^{2+} free. #Predicted response (8K + 100Na) if the individual effects of 8K and 100Na were additive (i.e. $88.7\% \times 96.0\%$).

pathways, attenuates fatigue (Cairns *et al.* 1998; Zhao *et al.* 2005; Germinario *et al.* 2008). These combined findings provide support for the $[\text{Ca}^{2+}]_{\text{t-sys}}$ depletion proposal. However, pre-equilibration at different $[\text{Ca}^{2+}]_o$ also influences $[\text{Ca}^{2+}]_i$ (Naro *et al.* 2003) making it plausible that nominally Ca^{2+} -free solutions exacerbate fatigue by lowering subsarcolemmal $[\text{Ca}^{2+}]_i$ during stimulation (Stroffekova & Heiny, 1997). In fact, experimentally lowering the resting $[\text{Ca}^{2+}]_i$ contributes to impaired excitability during fatigue in amphibian muscle (Usher-Smith *et al.* 2006). Nevertheless, resting $[\text{Ca}^{2+}]_i$ normally increases, alongside Ca^{2+} fluxes between the SR and myoplasm, during fatigue with repeated tetani (Allen *et al.* 2008), but just how subsarcolemmal $[\text{Ca}^{2+}]$ changes remains unknown.

The manipulation of fatigue kinetics with various $[\text{Ca}^{2+}]_o$ could involve a Ca^{2+} – K^+ interaction since elevated $[\text{Ca}^{2+}]_o$ restores force in K^+ -depressed muscle (Cairns *et al.* 1998). This effect is linked to a small repolarization, possibly due to increased Ca^{2+} -activated K^+ conductance (Jacquemonod & Allard, 1998), which is sufficient to account for the entire force recovery according to the peak tetanic force–resting E_m relationship (Cairns *et al.* 1998). However, a more likely physiological scenario involves lowered $[\text{Ca}^{2+}]_o$ acting in concert with diminished Na^+ and K^+ gradients (Ca^{2+} – Na^+ – K^+ interaction). Figure 3 shows the synergistic depressive effects of very low $[\text{Ca}^{2+}]_o$ superimposed on the Na^+ – K^+ interaction, whilst markedly raised $[\text{Ca}^{2+}]_o$ restored force. This force modulation by $[\text{Ca}^{2+}]_o$ may involve variations of E_m (Cairns *et al.* 1998), other processes that influence excitability (Usher-Smith *et al.* 2006), or the voltage sensors (Balog & Fitts, 2001). Despite these proposals a role for transsarcolemmal $[\text{Ca}^{2+}]$ remains speculative until $[\text{Ca}^{2+}]_{\text{t-sys}}$ and/or subsarcolemmal $[\text{Ca}^{2+}]_i$ have been measured during fatigue.

Chloride. When $[\text{Cl}^-]_o$ is reduced from 127 to 10 mM in non-fatigued muscle (to determine what normal $[\text{Cl}^-]_o$ does rather than to mimic a fatigue-induced change), force is initially depressed ($\sim 10\%$) but subsequently recovers fully (Cairns *et al.* 2004) when Cl^- efflux repolarizes the E_m (Hodgkin & Horowicz, 1959; McCaig & Leader, 1984). Thus, extreme reductions of $[\text{Cl}^-]_o$ *per se* would cause little fatigue. However, with pre-equilibration at low $[\text{Cl}^-]_o$ the ensuing fatigue is initially more rapid during either continuous or repeated tetani in slow- and fast-twitch muscle (Birnberger & Klepzig, 1979; Cairns & Dulhunty, 1995; Cairns *et al.* 2004) but the final force loss did not differ (Cairns *et al.* 2004). Similar fatigue responses occur when g_{Cl} is reduced (De Luca *et al.* 1990; van Lunteren *et al.* 2007). Also, twitch depression during repeated stimulation in skinned fibres is faster in Cl^- -free conditions (Dutka *et al.* 2008). These combined

observations at low $[\text{Cl}^-]$ may be caused by greater K^+ efflux (Cairns & Dulhunty, 1995; Dutka *et al.* 2008) and/or greater K^+ -induced depolarization (Dulhunty, 1979; Cairns *et al.* 2004). Hence, a normal $[\text{Cl}^-]_o$, $[\text{Cl}^-]_i$ and g_{Cl} prior to exercise provides resistance against fatigue.

The possibility of a Cl^- – K^+ interaction has been examined with two approaches. First, in muscles pre-equilibrated at low $[\text{Cl}^-]_o$ the subsequent tetanus depression at raised $[\text{K}^+]_o$ was more rapid and extensive than at normal $[\text{Cl}^-]_o$ (Cairns *et al.* 2004). Similar effects occur with raised $[\text{K}^+]_o$ in myotonic muscle, which has a reduced g_{Cl} (Birnberger & Klepzig, 1979). This effect is linked to greater depolarization (Cairns *et al.* 2004) as predicted from the GHK equation (Cairns *et al.* 1997). Hence, low $[\text{Cl}^-]_o$ caused movement down the peak tetanic force–resting E_m relationship with additional force loss through greater K^+ -induced depolarization. Second, in muscles pre-equilibrated at raised $[\text{K}^+]_o$, the subsequent lowering of $[\text{Cl}^-]_o$ by one-half or pharmacological blockade of g_{Cl} restored both force and excitability (Pedersen *et al.* 2005). This recovery occurred without altering E_m (Pedersen *et al.* 2005) and may involve smaller inhibitory currents so that a diminished I_{Na} is still sufficient to maintain excitability (Nielsen *et al.* 2004; Pedersen *et al.* 2004, 2005). These apparently conflicting findings may not be mutually exclusive but depend on the different experimental protocols. Possibly having a normal $[\text{Cl}^-]_o$ and normal g_{Cl} before exercise is protective, but when muscles are first depolarized, a lowered $[\text{Cl}^-]_o$ or lowered g_{Cl} then becomes protective. This idea is also consistent with the biphasic contractile response seen during repetitive muscle activity in severely myotonic patients (Ricker *et al.* 1978).

Hydrogen. A decreased pH_i (increased $[\text{H}^+]_i$) associated with lactic acid accumulation and $[\text{K}^+]_i$ depletion (Lindinger & Heigenhauser, 1991) has long been regarded as a major player in fatigue (Fitts, 1994) but this hypothesis is now questioned since a large acidosis has little detrimental effect on maximum force at body temperature (Cairns, 2006; Allen *et al.* 2008). Neither a preconditioning extracellular acidosis (Mainwood & Cechetto, 1980; Kristensen *et al.* 2005) nor alkalosis with raised $[\text{HCO}_3^-]_o$ (Lindinger *et al.* 1990; Broch-Lips *et al.* 2007) has any influence on fatigue resistance in isolated muscles. However, a H^+ – K^+ interaction has been demonstrated since acidosis partially restores force at raised $[\text{K}^+]_o$ in fibre-bundles from human patients (Lehmann-Horn *et al.* 1987) and in amphibian muscle (Renaud & Light, 1992). The notable observation that lactic acid restores K^+ -depressed force in rat soleus muscle (Nielsen *et al.* 2001) has been confirmed in other fast- and slow-twitch preparations (Pedersen *et al.* 2004, 2005; Hansen *et al.* 2005; Kristensen *et al.* 2005). Acidosis

restores excitability in K^+ -depressed muscle (Nielsen *et al.* 2001; Pedersen *et al.* 2004, 2005) without altering E_m (Lehmann-Horn *et al.* 1987; Pedersen *et al.* 2005), but acidosis also has small effects on action potentials in normal solutions (Pedersen *et al.* 2004, 2005). However, acidosis has no effect on charge movement in depolarized amphibian fibres (Balog & Fitts, 2001). Clearly acidosis shifts the peak tetanic force–resting E_m relationship rightwards towards more depolarized E_m , so that muscles can withstand a 2–3 mM higher $[K^+]_o$ although acidosis does not abolish depressive K^+ effects.

The precise mechanism by which acidosis restores excitability may involve two separate mechanisms with intracellular H^+ effects mediated via inhibition of g_{Cl} (Pedersen *et al.* 2004, 2005; Bennetts *et al.* 2007) and extracellular H^+ effects via reducing inactivation of voltage-dependent Na^+ channels (Lehmann-Horn *et al.* 1987). In support of the first mechanism, a three-way H^+ – K^+ – Cl^- interaction is illustrated in skinned rat fast-twitch fibres using low $[K^+]_i$ under acidotic conditions in normal and Cl^- -free conditions (Pedersen *et al.* 2005). An intracellular acidosis allows greater force production in K^+ -depolarized fibres at normal Cl^- ; however, in Cl^- -free conditions the protective effect of acidosis was not evident. If t-tubular g_{Cl} is reduced by acidosis, it becomes likely that a reduced I_{Na} will still trigger action potentials during trains (Pedersen *et al.* 2005). The second mechanism is shown when acidosis causes an increased I_{Na} in voltage clamp depolarized fibres (Lehmann-Horn *et al.* 1987).

Summary. Changes of transsarcolemmal K^+ , Na^+ , Ca^{2+} , Cl^- and H^+ gradients exert interactive effects on muscle force and need to be considered together. A lowered K^+ gradient appears to be the central player around which lowered Na^+ , Ca^{2+} and Cl^- gradients act synergistically to contribute to fatigue, whereas raised $[H^+]_o$, $[H^+]_i$, $[Ca^{2+}]_o$ (or reduced $[Cl^-]_o$ or g_{Cl} during late fatigue) may combat depressive K^+ effects to delay or attenuate fatigue (Fig. 4).

Ion-metabolite interactions and muscle function

Several metabolic changes occur in muscle alongside ion shifts during intense exercise. These include lowered $[ATP]$, lowered $[glucose]$ or glycogen content, and elevated $[H_2PO_4^-]$ and $[lactate^-]$ (Lindinger *et al.* 1987; Fitts, 1994; Allen *et al.* 2008). The question now arises whether any of these metabolic changes can amplify or dampen the effects of ionic changes during fatigue. In principle, metabolites could exacerbate ion shifts, modulate ion channel conductances, or influence ion-sensitive processes such as sarcolemmal excitability. Indeed, observations supporting such inter-

actions include the following: metabolically exhausted fibres display a markedly increased K^+ conductance (Fink & Lüttgau, 1976); pharmacologically inhibited mitochondrial function is linked to greater depolarization and impaired t-tubular membrane excitability (Ørtenblad & Stephenson, 2003); and lowered energy consumption (by inhibiting myosinATPase activity and SR Ca^{2+} release) postpones the decline of excitability seen during prolonged tetani (Macdonald *et al.* 2007). These effects on excitability may all relate to $[ATP]_i$ depletion and/or altered carbohydrate metabolism, especially in subsarcolemmal microdomains (Dutka & Lamb, 2007b).

ATP. A lowered $[ATP]_i$ may: (i) reduce Na^+ – K^+ -pump activity, thereby accelerating rundown of K^+ and Na^+ gradients along with greater depolarization (Clausen, 2003; Dutka & Lamb, 2007b); (ii) activate K_{ATP} channels to increase membrane conductance and exacerbate K^+ efflux (Fink & Lüttgau, 1976) (opening K_{ATP} channels with pinacidil accelerates both fatigue and the decline of excitability during repeated tetani; Gong *et al.* 2003; Kristensen *et al.* 2006); and (iii) combine with acidosis (pH 6.2) to alter the voltage dependence of the sarcolemmal $ClC-1$ channel by causing cooperative inhibition to decrease g_{Cl} (Bennetts *et al.* 2007).

Carbohydrate metabolism. Several studies demonstrate the importance of carbohydrate availability to maintain excitability. Depolarization-induced activation of excitation–contraction coupling is better maintained in fibres with higher glycogen content (Barnes *et al.* 2001), although lowered glycogen is not always detrimental (Goodman *et al.* 2005). Inhibition of glycolysis with iodoacetate reduces action potential amplitude (Fink & Lüttgau, 1976; Jones & Bigland-Ritchie, 1986). Also, glycogen phosphorylase-deficient patients (McArdle's disease) have an earlier decline of the M-wave during fatigue (Cooper *et al.* 1989). In keeping with these findings, glucose administration partially restored both the M-wave and force during late fatigue in muscles stimulated *in situ* (Karelis *et al.* 2003, 2005) and in exercising humans (Stewart *et al.* 2007). Such effects appear linked to a lesser depolarization during fatigue (Karelis *et al.* 2005), and are independent of insulin effects on the Na^+ – K^+ -pump (Karelis *et al.* 2003). Interestingly, glucose infusion is also a clinical treatment for hyperkalaemia (Stokes, 1989). These results support the notion that glycogenolytic/glycolytic fluxes maintain triadic $[ATP]_i$ for the Na^+ – K^+ -pump (Dutka & Lamb, 2007b) and/or provide $lactate^-$ to attenuate fatigue (Karelis *et al.* 2004). Thus hypoglycaemia or muscle glycogen depletion may amplify the effects mediated by K^+ -induced depolarization to reduce excitability during exercise.

Reactive oxygen species. Another effect of increased metabolic activity in working muscle is the production of reactive oxygen species (ROS) (Allen *et al.* 2008), which can induce a 10 mV depolarization and reduce action potential amplitude (Edwards *et al.* 2007; van der Poel *et al.* 2007). Moreover, *N*-acetylcysteine, an antioxidant used to combat ROS, attenuates the decline of $\text{Na}^+\text{--K}^+$ -pump activity and increases the time to exhaustion during prolonged intense exercise (McKenna *et al.* 2006). These findings lead to the intriguing possibility that effects of ROS may interact with a diminished K^+ gradient to further impair contraction during exercise.

Summary. An exercise-induced decline of $[\text{ATP}]_i$ (Fitts, 1994), muscle glycogen depletion or glucose depletion

(Ahlborg *et al.* 1967; Fitts, 1994) and/or elevation of ROS (Allen *et al.* 2008) may all contribute to fatigue by exacerbating the effects of a diminished K^+ gradient (Fig. 4).

Integrative perspective: interactive contributions of ions to fatigue

With respect to the functional design of skeletal muscle, it makes sense to have an operational range over which appreciable ionic and metabolic changes occur prior to any force decline, otherwise exercise would terminate rapidly. This safety margin concept is well supported by the large changes of $[\text{Na}^+]_o$, $[\text{K}^+]_o$, or depolarization needed to markedly reduce force (Fig. 2A and C).

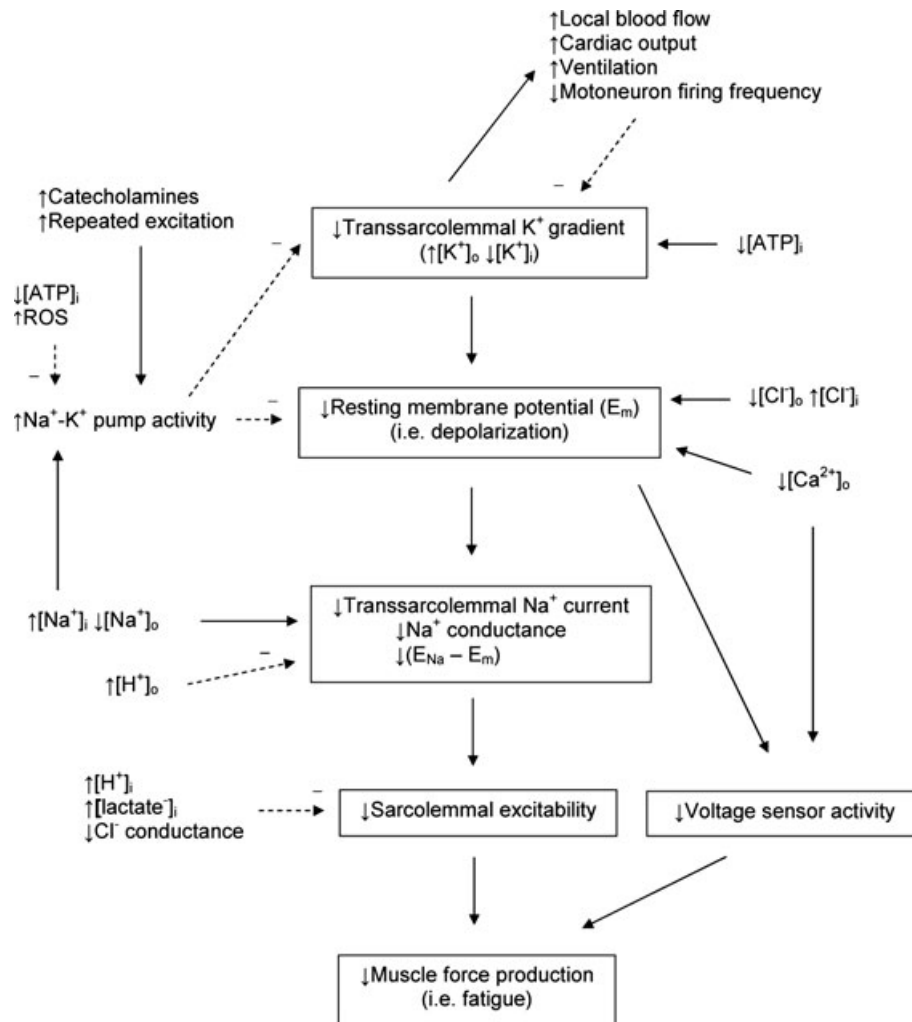


Figure 4. Model to illustrate potential integrated physiological mechanisms by which ion-ion interactions, ion-metabolite interactions and catecholamines influence muscle force production during fatiguing exercise

Continuous lines with arrow indicate support for the subsequent process. Dashed lines with arrow indicate resistance to the following process.

Moderate-intensity exercise. The ion gradients show moderate perturbations during submaximal work with smaller $[K^+]$ changes, and minor Na^+ , Ca^{2+} and Cl^- shifts; H^+ and lactate $^-$ may be unchanged (Ahlborg *et al.* 1967; Fitts, 1994; Sejersted & Sjøgaard, 2000; McKenna *et al.* 2006). This arises because of fewer action potentials, longer rest periods between contractions, adequate perfusion, and K^+ released by working fibres being sequestered into quiescent (non-recruited) fibres (Lindinger *et al.* 1995; Sjøgaard & McComas, 1995). Even when greater ionic changes occur there are several features existing which support a safety margin: the E_m may not be depolarized excessively because of excitation-induced stimulation of Na^+-K^+ -pumps (Hicks & McComas, 1989; Clausen, 2003; Lindinger *et al.* 2005) and/or membrane stabilization with Cl^- (Dulhunty, 1979; Cairns *et al.* 2004) or Ca^{2+} (Stokes, 1989; Cairns *et al.* 1998); action potential amplitude can fall considerably before any force loss (Cairns *et al.* 2003); and some action potential skipping (intermittent failure to generate action potentials during train stimulation) is either not detrimental, or may even increase force, in muscle depressed by raised $[K^+]_o$, lowered $[Na^+]_o$ or fatiguing stimulation (Cairns *et al.* 1997, 1998, 2003).

Moreover, raised $[K^+]_i$ stimulates several physiological processes that are likely to combat fatigue and benefit recovery. Importantly, submaximal contractions are potentiated at 7–10 mM $[K^+]_o$ in both fast- and slow-twitch muscles (Fig. 2A, Gallant & Donaldson, 1989; Renaud & Light, 1992; Gutierrez *et al.* 1996). This may involve an increased sensitivity of the voltage sensors (Chua & Dulhunty, 1988; Gallant & Donaldson, 1989) or increased resting $[Ca^{2+}]_i$ (Jacquemonde & Allard, 1998), which ensures that more Ca^{2+} released from the SR binds to troponin. Also, the blood flow to contracting muscle fibres is increased (Sjøgaard & McComas, 1995; Juel *et al.* 2007), through local vasculature effects and enhanced cardiovascular drive following stimulation of group III and IV afferents in the interstitium (Fig. 4, Jones & Bigland-Ritchie, 1986; Decherchi *et al.* 1998). This latter pathway may also stimulate ventilation and influence the CNS to reduce motoneuron firing frequency (Fig. 4, Jones & Bigland-Ritchie, 1986). These effects of raised $[K^+]_i$ should all assist whole-body exercise performance.

We suggest that with moderate-intensity exercise the combined ion shifts would usually be insufficient to cause fatigue. Rather the K^+ shifts would help maintain exercise through several integrated physiological mechanisms. Performance during very prolonged submaximal exercise may be limited by impaired motor drive (Presland *et al.* 2005), severe depletion of blood or muscle carbohydrate (Ahlborg *et al.* 1967; Fitts, 1994; Allen *et al.* 2008) or production of ROS (McKenna *et al.* 2006; Allen *et al.* 2008). However, the latter two mechanisms may contribute to fatigue, in part, through interactions with the smaller decline of K^+ gradient.

High-intensity exercise. Ionic shifts are much more extensive during heavy exercise (Table 1) than submaximal work. However, some protection is conveyed to working muscle through elevated levels of catecholamines (adrenaline/noradrenaline) (Clausen, 2003; Lindinger *et al.* 2005). These hormones, or β -adrenergic agonists, transiently restore force in K^+ -depressed muscle (Cairns *et al.* 1995; Hansen *et al.* 2005; de Paoli *et al.* 2007), slow the inhibitory effects of raised $[K^+]_o$ when combined (Clausen & Everts, 1991), and resist fatigue during repeated or prolonged tetani (Juel, 1988a; Cairns & Dulhunty, 1994; Clausen & Nielsen, 2007). These effects are thought to be mediated by enhanced Na^+-K^+ -pump activity (Fig. 4), which reduces depolarization to cause movement up the peak tetanic force–resting E_m relationship (Fig. 2C), and increases the Na^+ gradient (Cairns *et al.* 1995; Clausen, 2003; Hansen *et al.* 2005). In addition, the markedly elevated $[H^+]_i$ and $[H^+]_i$ may delay excessive fatigue. Nevertheless, we suggest that with increasing workload (and duration) a point is reached where the combined ionic changes in various compartments (Table 1, Fig. 1), are large enough to exceed the safety margin so that fatigue ensues.

The possible mechanisms by which larger ionic changes impact muscle performance are illustrated in Fig. 4. Multiple ion–ion interactions occur with K^+ being the central player about which interactive effects can be manifested. We suggest that during intense dynamic exercise of 5–20 min duration the combined rundown of transsarcolemmal K^+ , Na^+ and Cl^- gradients (and potentially lowered $[Ca^{2+}]_{t-sys}$ or subsarcolemmal $[Ca^{2+}]_i$) indeed contribute to severe fatigue. A key interaction with K^+ is likely to involve a reduced Na^+ gradient, which shifts the peak tetanic force–resting E_m relationship leftwards so that force loss occurs with less depolarization. Also interactions with diminished $[ATP]_i$, glucose or glycogen, and elevated ROS may reduce force, especially in fast-twitch fibres. Tempering these comments we note that during brief high-intensity exercise (e.g. 30 s all-out bout) there is less time for large ion concentration changes, with $[Na^+]_i$ and $[Cl^-]_i$ showing only non-significant increases (Kowalchuk *et al.* 1988). In this case, the role of ion–ion interactions is likely to be diminished, with raised inorganic phosphate ($H_2PO_4^-$) and lowered $[ATP]_i$ likely to be key factors limiting performance (Karatzafieri *et al.* 2001). Clearly, the heavy work needs to be of sufficient duration for multiple ionic changes to occur. Moreover, we very much support the notion that severe fatigue is both task dependent and multifactorial (Enoka & Stuart, 1992; Cairns *et al.* 2005; Allen *et al.* 2008), but during intense exercise of 5–20 min duration we regard ionic interactions to be key factors in the force loss. We now propose a modified K^+ hypothesis of fatigue: during high-intensity exercise a rundown of the transsarcolemmal K^+ gradient is the dominant cellular

process around which interactions with other ions and metabolites occur, thereby contributing to fatigue.

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